Mason Fleming

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EDUCATION

Purdue University, — **West Lafayette, IN**
Bachelor of Science, Computer Engineering — **May 2026**
Concentrations/Minor: Microelectronics — **GPA: 3.1/4.0**

PROFESSIONAL EXPERIENCE

Purdue University School of Electrical and Computer Engineering — **West Lafayette, IN**
Undergraduate SCALE Researcher (Dr. Peter Bermel) — **January 2024 – Present**

- Created Synopsys Sentaurus TCAD and Geant4 simulations to analyze transistor architectures' particle interactions and transient current response in radiation environments.
- Compared models to experimental data, reducing drain current perturbation by ~12% and identifying sub-4 nm reliability tradeoffs.
- Designed embedded radiation testing system (Arduino, ESP32, custom PCBs) with synchronized fault logging (GPIO, UART, shift registers) for system validation.
- Capturing SEU events (10^6–10^7 particles/cm^2) and correlated hardware faults with simulation outputs for system level validation.

University of Texas at Austin Cockrell School of Engineering — **Austin, TX**
Undergraduate Research Intern (Dr. David DiCarlo) — **June 2023 – August 2023**

- Designed data acquisition and control system (Permedia scripting, motion control, sand-bed hardware) for 2D flow studies.
- Built synchronized system (motion → fluid injection → optical imaging) enabling reproducible experiments with precise timing .
- Used MATLAB image processing to generate saturation maps and quantify plume migration (~30 mL breakthrough).
- Measured ~66% trapping efficiency, demonstrating improved vertical containment in fine-grained systems.

LEADERSHIP EXPERIENCE

Purdue IEEE EMBS — **West Lafayette, IN**
Club Chair — **May 2025 – Present**

- Expanded membership from 3 to ~20 through outreach, collaborations, and structured technical project meetings.
- Developed onboarding curriculum (Arduino, firmware, Git, PCB design, soldering) enabling independent project execution.
- Led development of biomedical systems that capture neurodegenerative features for diagnosis and monitoring.
- Organized EMG prosthetic claw workshop where myoelectric signals were used to control the contraction of a 3D-printed claw.

PROJECT EXPERIENCE

Live Street Parking Detection System (Senior Design) — **West Lafayette, IN**
Systems Engineer (Edge-to-Cloud) — **Jan. 2026 – Present**

- Built cellular module with solar charging for remote image transmission (Arducam, Raspberry Pi), including current spike protection.
- Designed distributed edge-to-cloud system (Raspberry Pi, SSH, LTE, Solar) for live image capture, transmission, and processing under latency and bandwidth constraints.
- Developed backend (FastAPI/Flask, Nginx, systemd) enabling concurrent uploads for fault-tolerant, reliable processing.
- Built frontend (HTML/CSS/JS) with API-driven architecture enabling real time visualization and low latency user interaction.

Multimodal Embedded Health Systems (Purdue IEEE EMBS) — **West Lafayette, IN**
Systems Engineer (Embedded & ML), Projects Lead — **Aug. 2025 – Present**

- Developed embedded system (Arduino/ESP32, Raspberry Pi, I²C/SPI/UART) for synchronized multimodal sensor data acquisition.
- Designed end-to-end pipeline (analog front-end → wireless transmission → edge inference) optimizing signal integrity, system performance, and data reliability.
- Built ML pipeline (NumPy, SciPy, XGBoost, CNN/LSTM) for clinical feature extraction with validation and degraded-input fallback.

GP2040 Analog Arcade Controller — **West Lafayette, IN**
Embedded Systems Engineer — **Jun. 2025 – Aug. 2025**

- Designed embedded system (RP2040 firmware, ADS1219, I²C) with high-resolution analog input and low latency USB HID output.
- Extended digital firmware to support analog inputs while maintaining GPIO compatibility for analog joystick support.
- Optimized PCB design and signal conditioning (RC filters, voltage dividers) to reduce noise and improve signal integrity .

ASIC Communication System (TX/RX/Data Buffer) — **West Lafayette, IN**
RTL Design Engineer — **Nov. 2025 – Dec. 2025**

- Designed RX module (Verilog/SystemVerilog, Xilinx Vivado, QuestaSim) ensuring integration with TX, buffer, and AHB interface.
- Developed modular RTL architecture and Git workflow improving debugging efficiency and reducing integration risk.
- Led system integration under incomplete team contribution, delivering functional design on schedule.

LICENSURE AND CERTIFICATIONS

- Semiconductors and Microelectronics Certificate - to be issued by Purdue University — **May 2026**

RELEVANT SKILLS

- C, C++, Python, MATLAB, RISC-V, Arduino, ESP32, RP2040, PCB Design, I²C, SPI, UART, Sentaurus TCAD, Geant4, Xilinx Vivado, QuestaSim, FastAPI, Flask, Nginx, Git, Make, NumPy, SciPy, XGBoost, CNN/LSTM, KiCad, SolidWorks, Soldering